MATLAB Reinforcement Learning Onramp : Agent Training & Learning Curve

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Task 1 : Course Completion

Completed 100% Reinforcement Learning Onramp course and obtained the course completion certificate.



Task 2 : RL Agent Training

Run the training example in MATLAB Onramp. Capture the Training Progress window and Learning Curve after execution.

Observation Table

Parameter	Observation
RL Environment	Dice Game (targetDice Simulink Environment)
Agent Used	rlQAgent (Q-Learning Agent)
Number of Training Episodes	2000
Initial Performance	Around -20 episode reward
Final Performance	Episode Reward = -3
Training Time	00:16:27
Training Stopped / Converged At	Episode 2000

Screenshot 1 : Training Progress



Screenshot 2 : Learning Curve



Questions :

20. What type of RL environment is used in the Onramp exercise?

The RL environment used is the Dice Game (targetDice) Simulink environment.

21. What type of agent is trained?

The trained agent is an rlQAgent (Q-Learning Agent).

22. What is the purpose of training the agent?

The purpose is to learn an optimal policy that selects actions to maximize the cumulative reward in the Dice Game.

23. What performance measure is used during training?

The main performance measures are Episode Reward and Average Reward.

24. How does the agent's performance change as training progresses?

Initially, the rewards are low because the agent explores randomly. As training continues, the rewards improve and become more stable, showing that the agent is learning.

Task 3 : Learning Curve Analysis

Training Stage	Episode Range	Reward/Performance	Observation
Initial	1-100	-20 to -10	Random actions, very poor performance
Early Training	101-500	-10 to -6	Reward begins improving
Middle Training	501-1000	-7 to -5	Agent learns better decisions
Late Training	1001-1800	-5 to -4	Performance becomes more stable
Final	1081-2000	Around -3	Training converges with consistent rewards

Analysis Questions :

25. What does the learning curve represent?

The learning curve represents how the episode reward changes over the training episodes, showing the agent's learning progress.

26. Why is the reward generally low during the initial training episodes?

Because the agent starts without prior knowledge and learns by exploring different actions, which results in lower rewards initially.

27. What indicates that the agent is learning?

A gradual improvement in episode rewards and a more stable average reward indicate that the agent is learning.

28. Does the reward increase consistently throughout training? Explain.

No. The reward fluctuates because the agent balances exploration and exploitation, but the overall trend shows improvement.

29. How can you identify convergence from the learning curve?

Convergence is identified when the reward values become relatively stable and show little long-term improvement. In this experiment, training finished after 2000 episodes.

30. What could cause fluctuations in the learning curve?

Fluctuations are caused by random exploration of actions, different state-action experiences and stochastic nature of the environment.

31. What happens if the agent is trained for a larger number of episodes?

The agent may improve slightly, but after convergence the performance changes very little and becomes stable.

32. Why is it important to analyze the learning curve rather than considering only the final reward?

The learning curve shows the complete learning behavior including initial performance, improvement, fluctuations, stabilization, and convergence whereas the final reward shows only the last outcome.